

Oscar A. Flores Gaitán

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Education

Ph.D. in Astronomy & Astrophysics

Department of Physics & Astronomy, University of Delaware

Newark, DE

Aug 2026 – Present

- Advisor: Prof. Sarah Dodson-Robinson

Licentiate in Physics

Department of Physics, Universidad del Valle de Guatemala

Guatemala City

Jan 2022 – May 2026

- Honors: Magna Cum Laude
- Thesis: "*Decoding the Barnard's Star Planetary System: A Time Series Based Analysis of the Planetary Architecture*" (Advisor: Prof. Sarah Dodson-Robinson)

Research Interests

- Precision Doppler detection and characterization of exoplanetary systems.
- Modeling and mitigating stellar activity in radial velocity and spectroscopic time series.
- Developing and applying machine learning and statistical techniques to astronomical data.

Publications

First-Author

1. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V.. "*How many terrestrial planets orbit Barnard's Star?*". [Manuscript in preparation, 2026].

Co-Authored & Contributor

2. Zhao, L.L. et al., **including Flores Gaitán, O.A.**. "*The Extreme Stellar-Signals Project IV*". [Manuscript in preparation, 2026].
1. Dressing, C.D., Adams, D., Alecian, E. et al., **including Flores Gaitán, O.A.**. "*Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory*". arXiv preprint (2026). [arXiv:2608.11294](#) [🔗](#).

Software

1. **Flores Gaitán, O.A.**, & Hopper, S.. "*orbcloud: 3D orbital probability clouds from simulated exoplanet posteriors*". Zenodo (2026). DOI: [10.5281/zenodo.20935725](#) [🔗](#).

Consortia & Collaborations

- **Extreme Stellar Signals Project (ESSP)**: Collaboration Member through the Sidera Research Group, University of Delaware.

Talks & Posters

6. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., Rubio-Herrera, E., "*How many terrestrial planets orbit Barnard's Star?*", **Contributed Talk** at APS Mid-Atlantic Section Meeting 2025, Penn State Harrisburg, Middletown, PA, November 15, 2025. [Link](#) [🔗](#)
5. **Flores Gaitán, O.A.**, "*Hunting for Exoplanets: Real Planets or Stellar Processes?*", **Invited Talk** at Universidad del Valle de Guatemala's Physics Week, Guatemala City, October 25, 2025.
4. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., "*Disentangling Planetary and Stellar Signals in Barnard's Star*", **Poster** at Physics Science Fair at Universidad del Valle de Guatemala's Physics Week, Guatemala City, October 24, 2025.
3. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., "*Disentangling Planetary and Stellar Signals in Barnard's Star*", **Virtual Poster** at NASA's Towards the Habitable

Worlds Observatory: Visionary Science and Transformational Technology, Johns Hopkins University, Washington, DC, July 28-31, 2025.

2. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., "*Disentangling Planetary and Stellar Signals in Barnard's Star*", **Virtual Poster** at Sagan Summer Workshop, NASA Exoplanet Science Institute, California Institute of Technology, Pasadena, CA, July 23, 2025.
[Link](#) 

1. **Flores Gaitán, O.A.**, "*Detecting and Characterizing Exoplanets with TESS*", **Invited Talk** at Universidad del Valle de Guatemala's Astronomy Club, Guatemala City, May 9, 2025.

Research Experience

Graduate Researcher

Department of Physics & Astronomy, University of Delaware

Newark, DE

Aug 2026 – Present

Advisor: Prof. Sarah Dodson-Robinson

- Analyzing solar observations to test and calibrate spectral coherence methods for measuring stellar rotation and magnetic activity cycles.
- Developing dual-frequency coherence methods to disentangle stellar activity from planetary signals in radial velocity (RV) datasets.

Undergraduate Visiting Researcher

Department of Physics & Astronomy, University of Delaware

Newark, DE

Jun 2025 – Jul 2026

Advisor: Prof. Sarah Dodson-Robinson

- Evaluated advanced statistical methods to distinguish genuine exoplanetary signals from stellar noise in high-precision RV data.
- Implemented gaussian process (GP) regression and frequency-domain analysis to mitigate stellar variability and improve the reliability of RV detections.
- Analyzed 25 years of multi-instrument RV datasets to determine the planetary architecture of Barnard's Star, utilizing advanced statistical metrics for model validation.

Undergraduate Researcher

Department of Physics, Universidad del Valle de Guatemala

Guatemala City

Nov 2024 – May 2025

Advisor: Prof. Eduardo Rubio-Herrera

- Processed and analyzed photometric time series data from the TESS mission to identify exoplanet candidates.
- Developed an algorithm to detect transits and extract orbital parameters from stellar light curves.

Academic Appointments & Teaching

Graduate Teaching Assistant

Department of Physics & Astronomy, University of Delaware

Newark, DE

Aug 2026 – Present

- 4 course sections: PHYS 221 (Introductory Physics Laboratory I).

Undergraduate Teaching Assistant & Laboratory Assistant

Physics Department, Universidad del Valle de Guatemala

Guatemala City

Jan 2024 – May 2025

- 7 course sections: Physics I (FF2016), Physics II (FF2017), Physics III (FF2018), and Classical Mechanics I (FF3027).

Academic Tutor

Universidad del Valle de Guatemala

Guatemala City

Jun 2022 – Present

- Provided 350+ hours of individual tutoring for high school and university students in Calculus I–III, Differential Equations, and Physics I–III.

Industry & Professional Experience

Data & Research Analyst

Vekst Consulting Group

Guatemala City

Feb 2026 – May 2026

- Led scientific consulting and quantitative modeling projects, applying advanced statistical methods to financial and market data analysis.
- Transformed rigorous research into high-value strategic assets and technical documentation for the consulting sector.
- Designed, developed, and deployed the firm’s official website, expanding technical and commercial visibility.

Mentorship

University of Delaware Undergraduate Research Program

Newark, DE

Primarily advised by Prof. Sarah Dodson-Robinson

Jun 2026 – Present

- Student: Isabel Cambronero Jiménez (Universidad de Costa Rica).
- Project: Application of the Cepstrum Technique to Exoplanet Radial Velocity Time Series.

Workshops & Summer Schools

Cohere Labs Machine Learning Summer School

Remote

Cohere Labs

Aug 2026

Code/Astro Workshop

Santa Cruz, CA

University of California, Santa Cruz

Jun 2026

Alpha-Cen Astronomy Summer School

Remote

Alpha-Cen (Central American and Caribbean Astrophysics)

Aug 2025

Workshop on Astrophysics of High Energies and Cosmology

Guatemala City

Universidad Nacional Autónoma de México (UNAM)

Nov 2024

Fellowships, Grants & Awards

- **Code/Astro Full Financial Aid Grant**, University of California, Santa Cruz: Awarded full financial support for scientific software development workshop (2026).
- **Summer Visiting Research Fellowship**, Department of Physics & Astronomy, University of Delaware: Selected out of 147 international applicants (2025).
- **3x Distinguished Student Award**, Universidad del Valle de Guatemala: Awarded for top academic standing (2022, 2023, 2024).
- **Experimental Physics Record**, Universidad del Valle de Guatemala: Achieved the university’s historically most accurate experimental measurement of local gravitational acceleration ($g = 9.7876 \pm 0.0026 \text{ m/s}^2$) (2024).
- **Top 10 Finalist**, Guatemalan National Mathematics Olympiad (2020).

Technical Skills

Programming Languages: Python, L^AT_EX, R (Intermediate)

Data Analysis & Probabilistic Modeling: Statistical Inference, Gaussian Processes (GPs), Markov Chain Monte Carlo (MCMC), Bayesian Modeling, Time-Series Analysis

Machine Learning: Scikit-learn, XGBoost, Random Forests, Neural Networks, Autoencoders, K-Nearest Neighbors (KNN), Hyperparameter Tuning, Cross-Validation

Astronomy Tools: PyMC, celerite2, exoplanet, AstroPy, Lightcurve, Astroquery, ExoCTK, NWelch

Languages: Spanish (Native), English (C1), German (B2/C1)

Outreach & Leadership

- **Invited Panelist:** “Experiences of Internships Abroad for Aspiring Guatemalan Astronomers”, CAFE 2026 (Workshop on Astronomical Sciences: Education and Training), Universidad Rafael Landívar, Guatemala City (Jul 2026).
- **Science Fair Judge:** Evaluated undergraduate physics and engineering research projects at Uni-

versidad del Valle de Guatemala's Science Fair, Guatemala City (Jun 2026).

- **Workshop Developer & Instructor:** Designed and taught *"How to Apply to Internships in the US"* for undergraduate physics students in Latin America (Feb 2026).
- **Guest Lecturer & Teaching Assistant:** Collaborated on Extragalactic Astronomy lecture for high school students, Astro Antigua, Antigua Guatemala (Dec 2025).
- **Lead Organizer:** Physics Week 2025, a week-long conference and workshop series featuring national and international speakers, Universidad del Valle de Guatemala, Guatemala City (Oct 2025).
- **Workshop Developer & Instructor:** Designed and taught *"How to Write an Academic CV"* for first-year physics students, Universidad del Valle de Guatemala, Guatemala City (Oct 2025).
- **Active Member:** Physics Students Association (AEF-UVG), organizing public lectures and science outreach events (2025).
- **Author & Content Creator:** Authored undergraduate Physics laboratory guide and created educational laboratory videos at Universidad del Valle de Guatemala (2024). [Link](#) 
- **Founder & Manager:** *Countries Kingdom*, educational geography and linguistics outreach project with over **15,000 followers** and **5,000,000+ total views** (2021 – 2023). [Link](#) 

Media

3. **Actualidad UVG**, *"Oscar Flores Gaitán demonstrates the reach of Physics in the search for exoplanets (In Spanish)"*, **Feature Article** at Universidad del Valle de Guatemala, Guatemala City, 2026. [Link](#) 
2. **Ciencia UVG Podcast**, *"A UVG Physicist in Delaware (In Spanish)"*, **Podcast Episode** at Universidad del Valle de Guatemala, [YouTube \(Video\)](#)  / [Spotify \(Audio\)](#) , 2026.
1. **Educational Video Tutorial**, *"Measuring the gravitational acceleration g with a physical pendulum (In Spanish)"*, **YouTube Video** at Universidad del Valle de Guatemala, Guatemala City, 2024. [Link](#) 